

**Before the
Office of Science and Technology Policy
Washington, DC**

In the Matter of	)	
	)	
Request for Information on the Development	)	FR Docket No. 2025-02305
	)	
of an Artificial Intelligence (AI) Action Plan	)	

Reply Comments of Neha Ayyalapu

February 22, 2025



I am responding to the “Request for Information on the Development of an Artificial Intelligence (AI) Action Plan” issued jointly by the OSTP, NSF, NITRD, and NCO. One issue that must be addressed in AI policy development is the environmental costs associated with AI due to the vast computational power required to develop and deploy these systems. Energy demand for data centers in the United States has tripled in the last decade and is expected to nearly triple by 2030. A significant portion of this energy consumption comes from the growing demand for artificial intelligence tools, which is only expected to increase in the coming years. Current U.S. policy emphasizes advancing our AI capabilities and expanding infrastructure to keep pace with foreign competitors. While sustaining a competitive edge is important in today’s technological landscape, it is equally important to ensure that these efforts do not come at the expense of environmental sustainability.

I suggest two primary mechanisms for mitigating AI’s energy footprint while fostering responsible innovation. First, the administration should establish structured incentives to encourage the adoption of energy-efficient AI systems. Second, the government should implement standardized reporting methods and requirements for AI-related energy consumption. I expand on each of these below.

The first policy I propose is to offer tiered incentives for the use of more energy-efficient AI systems. This approach specifically targets three aspects of AI systems: (1) the training and deployment of the model itself, (2) the hardware involved in running AI models, and (3) the data centers that serve as infrastructure. Each of these areas presents significant opportunities for energy savings, and addressing them collectively will maximize impact. I briefly elaborate on the progress and potential for further energy efficiency in each of these verticals, supported by insights from computer science research.

The energy consumption of AI models is driven by two phases: training (the initial development of the model) and inference (when the model is actively used). Efforts to reduce energy consumption during training include Google's GLaM, an LLM that outperformed GPT-3 despite using 2.8 times less energy,¹ and techniques such as early stopping during hyperparameter optimization, which can reduce training energy consumption by up to 80%.² On the inference side, there are new systems such as CLOVER, which optimizes workloads through GPU resource partitioning and can reduce carbon emissions by up to 60% with minimal accuracy loss.³ Advancements in hardware also play a significant role in reducing the energy footprint of AI. For example, Google developed Tensor Processing Units (TPUs), hardware specialized for neural networks that improved carbon efficiency threefold⁴ while also performing operations over 15 times faster than some traditional CPUs and GPUs.⁵

Meanwhile, data centers, which serve as the backbone of AI infrastructure, present another significant opportunity for reducing environmental impact. Some solutions leverage power capping—dynamically adjusting power usage at a data center based on current workload demands—to prevent unnecessary energy waste.⁶ Recently, researchers at the University of Waterloo introduced an optimization to Linux network traffic processing, improving CPU cache

¹ Patterson, David, et al. "The Carbon Footprint of Machine Learning Training Will Plateau, Then Shrink." *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, vol. 55, no. 7, 2022, pp. 18-28.

² Foy, Kylie. "AI Models Are Devouring Energy. Tools to Reduce Consumption Are Here, If Data Centers Will Adopt." *MIT Lincoln Laboratory*, 22 Sep. 2023.

³ Li, Baolin, et al. "Clover: Toward Sustainable AI with Carbon-Aware Machine Learning Inference Service." *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC 2023)*, 2023, pp. 20:1–20:15.

⁴ Patterson, David, and Parthasarathy Ranganathan. "Designing sustainable AI: A deep dive into TPU efficiency and lifecycle emissions." *Google Cloud Blog*, 5 Feb 2025.

⁵ Sanmartín, Diego, and Vera Prohaska. "Exploration of TPUs for AI Applications" *Proceedings of the Second International Conference on Advances in Computing Research (ACR'24)*, vol. 956, 2024.

⁶ Chen, Huamin, and Sandro Mazziotto. "Introducing Climatix: Power capping AI applications for data center sustainability." *Red Hat Blog*, 12 Nov. 2024.

efficiency and potentially reducing data center energy consumption by up to 30%.⁷ All of these advancements highlight the potential to reduce the environmental footprint of AI— a potential that can be maximized through federal incentives and standards.

Thus, to accelerate progress and adoption of energy-efficient AI systems, I urge the administration to implement incentives modeled after successful programs in other industries. The government could introduce a tiered tax credit system for companies that develop or deploy AI models, hardware, or data centers that meet defined energy and sustainability benchmarks, with greater efficiency metrics corresponding to incrementally higher tax benefits. The federal government could establish a tiered Green AI Certification (ranging from Bronze to Platinum) to evaluate the environmental impact of AI systems, similar to the LEED rating system for buildings. This certification could assess additional factors such as hardware lifecycle sustainability and integration of renewable energy sources. Certified technologies could benefit from advantages in federal procurement opportunities or streamlined approval processes. To further drive innovation, the DOE and NSF could jointly administer a grant program to fund research in energy-efficient AI technologies, prioritizing high-impact areas identified by experts. While these are just a few proposals, any incentives implemented should promote incremental improvements as well as cutting-edge work across AI models, hardware, and data centers.

While incentivizing energy-efficient AI is important, it is challenging to measure progress without standardized metrics for performance and improvement. Although industry transparency has improved, the energy demands of AI are still often hidden by companies and invisible to users and policymakers. To address this, I introduce my second policy: the

⁷ Gooding, Matthew. "Changing Linux code could cut data center energy use by 30%, researchers claim." *Data Center Dynamics*, 22 Jan. 2025.

establishment of mandatory energy disclosure requirements for data centers and the development of energy transparency reporting guidelines for AI systems.

Following the European Union’s Energy Efficiency Directive,⁸ the U.S. should require data centers to report key performance indicators such as power usage effectiveness, water usage effectiveness, total energy consumption, and carbon emissions. Disclosure mandates would encourage data centers to adopt best practices for sustainability. Making these metrics publicly available would also enable policymakers, researchers, and consumers to better assess the environmental impact of AI infrastructure, compare efficiency across facilities, and push for improvements.

For AI models, I recommend that NIST and the DOE develop a standardized methodology for measuring the energy and carbon costs of AI training and inference across different model architectures and data center infrastructures. Once established, companies should be encouraged or incentivized to report the energy and carbon costs associated with training and operating their AI models. Public disclosure would increase market pressure for efficiency and reward energy-conscious AI development while enabling consumers and policymakers to make more informed decisions.

AI will continue to shape the future of the U.S. economy and society in the coming years, but its expansion must be balanced with sustainability efforts. By combining targeted incentives for energy efficiency with standardized reporting and transparency measures, the administration can ensure that AI development and innovation align with environmental responsibility.

⁸ European Commission. *Commission Delegated Regulation (EU) 2024/1364 of 14 March 2024 on the First Phase of the Establishment of a Common Union Rating Scheme for Data Centres*. Official Journal of the European Union, L 1364, 17 May 2024, pp. 1–15.